Mortality from the Influenza Pandemic of 1918-19: The Case of India

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Abstract:

Estimates of worldwide mortality from the influenza pandemic of 1918-19 vary widely, from 15 million to 100 million. In terms of loss of life, India was the focal point of this profound demographic event. In this paper, we calculate mortality from the influenza pandemic in India using panel data models and data from the *Census of India*. The new estimates suggest that, for the districts included in the sample, mortality was 11.41 million, compared with 15.46 million when calculated using the assumptions of Kingsley Davis (1951). We conclude that Davis' influential estimate of mortality from influenza in British India is overstated by a factor of over 35%. Future analyses of the effects of the pandemic on demographic change in India and worldwide will need to account for this significant downward revision.

Introduction:

The influenza pandemic of 1918-19 has been called the 'greatest medical holocaust in history' (Waring 1971, p.33) and the 'mother of all pandemics' (Taubenberger and Morens 2006). Research conducted in the 1920's indicates that global mortality during the pandemic exceeded 21.5 million (Jordan 1927, Johnson and Mueller 2002). More recent estimates have varied widely, suggesting that between 15 and 100 million people died in the short span of about a year (Patterson and Pyle 1991, Johnson and Mueller 2002). The country that faced the greatest devastation in terms of human mortality from influenza was India (Johnson and Mueller 2002, Table 3, p.112), then a British colony. Mortality estimates for India are also wide-ranging, from 12 to 20 million (*Census of India* 1921, Davis 1951, Mills 1986, Patterson and Pyle 1991). According to a recent estimate, mortality in India accounted for 38% of global mortality (Johnson and Mueller 2002). These figures are highly problematic, because "[d]eath totals for British India …. provide the largest single source of uncertainty for Asian and world mortality totals" (Patterson and Pyle 1991, p.18). Sound estimates of the impact of the pandemic are of critical importance to understanding demographic change in India and the world in the 20th century. The enormous numbers of lives lost and people sickened by the virus had consequences that lingered through successive generations long past the middle of the 20th century (Almond 2006; Johnson and Mueller 2002; Mazumder et. al. 2010; Mills 1986; Morens et. al. 2009).

Because of the unreliability of vital registration records of the time, the most influential study on this subject, Davis (1951), uses the data from the Indian census, collected in times of relative stability, as checks on preliminary mortality estimates based on death registration records. Using an average decadal population growth rate of 8% for the decades 1901-11 and 1921-31, he computes a shortfall of 18.5 million (Davis 1951, Appendix B, p. 237), which he attributes to the influenza pandemic. Mills challenges the assumption of 8% decadal growth before the pandemic, and instead uses a population growth rate of 6.8% for the period up to 1917 (Mills 1986, p.10). In this paper, we demonstrate that both sets of assumptions about growth rates before and after 1918-19 are inaccurate. Indeed, in addition to estimating the population growth leading up to 1918 and finding it to be lower than the 6.8% or 8% as posited in the above studies, we estimate the decadal population growth rate *after* the pandemic and find it to be well in excess of the 8% assumed earlier. This is in line with Klein's assertion about the inter-war period as "the beginning of a demographic revolution" (Klein 1990, p.33).

The aim of this paper, therefore, is to use analytic methods that were not available to earlier demographers to estimate mortality from the influenza pandemic of 1918-19 in British India using the census data. In doing so, we come up with a revised figure that is significantly lower than that calculated in the above studies.

Data

The data used in this paper come from six decennial censuses of British India, conducted in 1891, 1901, 1911, 1921, 1931, and 1941 (*Census of India*, various years). The 1891 census of India is the first one for which undercounting amounts to approximately 1% or less of the total population (Davis 1951, Table 7, p.27 and Appendix A, p.235 and Klein 1974, p.200). The choice of 1891 as the starting year is also informed by a concern for consistency --- the method of conducting the census changed between 1872 and 1881, and again between 1881 and 1891 (Davis (1951), Appendix A, p.235).

For the analysis, we focus on those districts of British India that were under direct rule. This ensures that populations for the districts included in the study were enumerated using a similar set of administrative procedures and comparable institutional structures. The districts belonging to the so-called princely states, which were not directly ruled, had their own civil service systems that were often not as well equipped to carry out the census as the directly ruled districts. The above data, therefore, form a panel of 199 districts (see Appendix 1 for a list), for which population is observed for all six censuses, for a total of 1,194 observations. The population statistics for each census were adjusted to conform to the district boundaries as of the 1941 census. This adjustment is discussed in the Census of 1941 (*Census of India*, 1941, various pages).

Methods:

A key contribution of this paper is to bring recently-developed methods in statistical demography that were not available to earlier historians and demographers to bear on the census data of British India to estimate mortality from the influenza pandemic of 1918-19. These methods improve our understanding of this important demographic phenomenon by simultaneously estimating population growth rates before and after the pandemic and the drop in population without any prior restrictions on what these should be. By using a panel of 199 districts, we bring to bear the full power of the census data for the estimation of the impact of the pandemic.

We fitted a series of panel data models to the census data that capture the average population growth across districts, while at the same time providing unique, model-based estimates of growth for each district. The general model can be expressed as

$$LPOP_{it} = \pi_{0i} + \pi_{1i}T_t + \pi_{2i}FLU_t + \pi_{3i}T_tFLU_t + \varepsilon_{it}$$

where $LPOP_{it}$ is the log of population in district i in year t, T_t is a time trend, FLU_t is a dummy variable defined as

$$FLU_t = \begin{cases} 0, & t \le 1918 \\ 1, & t > 1918 \end{cases}$$

and ε_{it} is a random error term. In the random coefficients version of the model, π_{0i}, π_{1i}, π_{2i}, and π_{3i} are the estimated random coefficients of the model, which consist of fixed terms γ_{00}, γ_{10}, γ_{20}, and γ_{30} respectively and four corresponding random (district-specific) terms with 0 mean. The random coefficients represent the district-specific values of the logarithm of population in 1891, the rate of population growth before the outbreak, the shift factor in the growth trajectory of population due to the influenza pandemic of 1918 (calculated at 1891), and the rate of population growth after the pandemic, providing a population model for each district (see Appendix 2 for details). For the equivalent fixed coefficients models, the coefficients π_{0i}, π_{1i}, π_{2i}, and π_{3i} in the above random coefficients model were replaced with the standard (fixed) coefficients β_{0i}, β_{1i}, β_{2i}, and β_{3i}.

All models were implemented and evaluated using the PROC MIXED and PROC PANEL modules in the SAS software (SAS 2011a,b). We conducted two tests to evaluate the different model specifications. The Hausman test indicated consistency of the random coefficients estimates, and the Breusch Pagan test indicated rejection of the fixed coefficients specification in favor of the random coefficients specification. While the results presented in Table 1, therefore, focus on the random coefficients models, mortality estimates from the equivalent fixed coefficients models are also presented for comparison, and demonstrate the robustness of the estimates across specifications.

In the first model, labeled the “unrestricted model” (Table 1, column 1), we allowed the growth rate before the influenza pandemic to differ from that after the pandemic, i.e., π_{3i} was included in the model. To compare this unrestricted model with the equivalent one using Davis' (1951) assumption of equal population growth rates before and after the decade of 1911-1921, we also analyzed a model which does not include a change in slope after the epidemic outbreak (i.e., π_{3i} is excluded). We label this model the “restricted (Davis) model” (Table 1, column 2). Table 1 also contains corresponding estimates of the number of deaths that occurred between 1918 and 1919 as a result of the epidemic and of annual rates of population growth before and after 1918-19.

[Table 1 about here]

A key difference between the unrestricted and restricted (Davis) models is the rates of population growth before and after 1918. The restricted (Davis) model, for which these two rates are restricted to be equal, shows a decadal rate of slightly above 8%. This is consistent with Davis (1951). However, as the coefficient estimates for the unrestricted model show, the assumption of equal growth rates before and after the pandemic is statistically untenable. The null hypothesis $H_0 : \pi_{3i} = 0, \forall i$ is rejected at the 1% level of significance, indicating that population grew at different rates before and after the pandemic. This is consistent with a key observation in Klein (1990).

To estimate the population drop from 1918 to 1919, we computed district growth trajectories to produce model-based estimates of the population change for each district. We then summed these changes across districts to create an estimate of the total population change. The population change was estimated to be 11.41 million people for the unrestricted model and 15.46 million people for the Davis model, showing that Davis' assumption yields an overestimate in the number of deaths in 1918-19 by over 4 million or 35%! Figure 1 is a graphical representation of the impact of relaxing the assumption of equal growth rates before and after the pandemic on the mortality estimates.

(Figure 1 about here)

As a robustness check, we estimated the same model excluding data from the 1891 census to eliminate the potential effects of the unusually devastating famines of the late 1890s on the models (Davis 1951, p.28, 39). These estimates (Columns 3 and 4 of Table 1) demonstrate that this only serves to exacerbate the difference between the mortality estimates for the unrestricted (13.04 million) and restricted (Davis, 23.36 million) models. Figure 2 illustrates the difference between the unrestricted and Davis models estimated without the 1891 census data. Finally, Table 1 also contains mortality estimates from the fixed coefficients models that correspond to the random coefficients models for which the results were originally reported. These estimates do not differ much from the random coefficients models.

(Figure 2 about here)

Discussion and Conclusion

The estimates from the above models suggest that influenza-related mortality in the districts of British India was in the vicinity of 11 million. This number is significantly lower than the 15 million estimated using the Davis model for the same set of districts. The reason for this difference lies in the assumption in Davis (1951) that the rate of population growth before and after the influenza pandemic was uniformly 8% per decade (Davis, 1951, Appendix B, p.237), an assumption that is statistically indefensible. The plots in Figures 1 and 2 compare the random coefficients models with and without Davis' restriction on population growth rates before and after the pandemic, showing the impact of relaxing the restriction on the estimate of mortality due to the pandemic. The unrestricted model shows a drop in population that is visibly smaller than that in the unrestricted model.

The findings of this paper raise a number of interesting questions, key among them being that of a possible connection between the influenza pandemic and the subsequent population growth spurt. According to Klein (1990), this spurt was the consequence of the gradual development of immunity to such diseases as malaria and plague. It is entirely plausible, however, that, as in the case of Iran, the influenza pandemic also played a role by causing a one-time spike in mortality among those who suffered from chronic malaria and, possibly, other diseases, thereby leaving a healthier but diminished population in its wake (Afkhami 2003). A second interesting and related question concerns the disproportionate mortality that influenza caused among people of childbearing age. This would have had a depressing effect on post-pandemic population growth. Taken together, the two above phenomena suggest that any future study of the impact of the influenza pandemic on India's population will need to disentangle two possibly countervailing effects.

In sum, using methods of statistical demographic analysis that were not available to earlier demographers who studied mortality from the 1918 influenza pandemic in India, we find evidence that suggests that the estimate that has stood for the past 60 years was overstated by over 35%. This finding is of great significance because it will necessitate a reappraisal of the impact of one of the most devastating epidemiologic events the world has seen in a country that, in terms of mortality, was the focal point of the event.

Appendix 1: List of districts used in the analysis

Name	Name	Name	Name	Name	Name
Agra	Budaun	Ganjam	Kanara	Muzaffarpur	Satara
Ahmedabad	Bulandshahr	Garhwal	Kangra	Mymensingh	Saugor
Ahmednagar	Buldana	Garo Hills	Karachi	Nadia	Shahabad
Ajmer Merwara	Burdwan	Gaya	Karnal	Nagpur	Shahjahanpur
Akola	Cachar	Ghazipur	Kheri	NainiTal	Shahpur
Aligarh	Calcutta	Godavari East	Khulna	Nasik	Sheikhupura
Allahabad	Cawnpore	Godavari West	Kistna	Nellore	Sholapur
Almora	Champaran	Gonda	Kolaba	Nilgiris	Sialkot
Ambala	Chanda	Gorakhpur	Koraput	Nimar	Sibsagar
Amraoti	Chhindwara	Gujranwala	Kurnool	Noakhali	Singhbhum
Amritsar	Chingleput	Gujrat	Lahore	North Arcot	Sitapur
Anantapur	Chittagong	Guntur	Lakhimpur	Nowgong	SouthArcot
Attock	Chittoor	Gurdaspur	Larkana	Pabna	South Kanara
Azamgarh	Coalpara	Gurgaon	Lucknow	Palamau	Sukkur
Bahraich	Coimbatore	Hamirpur	Ludhiana	24 Parganas	Sultanpur
Bakarganj	Cuddapah	Hardoi	Madras	Partabgarh	Surat
Balaghat	Cuttack	Hazaribagh	Madura	Pilibhit	Sylhet
Balasore	Dacca	Hissar	Mainpuri	Poona	Tanjore
Ballia	Darbhanga	Hooghly	Malabar	Puri	Thana
Banda	Darjeeling	Hoshangabad	Malda	Purnea	TharParkar
Bankura	Darrang	Hoshiarpur	Manbhum	Rae Bareli	Tinnevelly
BaraBanki	Dehra Dun	Howrah	Mandla	Raipur	Tippera
Bareilly	Dera Ghazi Khan	Hyderabad	Meerut	Rajshahi	Trichinopoly
Basti	Dharwar	Jalaun	Mianwali	Ramnad	Unao
Belgaum	Dinajpur	Jalpaiguri	Midnapore	Ranchi	Upper Sind Frontier
Bellary	Drug	Jaunpur	Mirzapur	Rangpur	Vizagapatam
Betul	East Khandesh	Jessore	Monghyr	Ratnagiri	Wardha
Bhagalpur	Etah	Jhang	Montgomery	Rawalpindi	West Khandesh
Bhandara	Etawah	Jhansi	Moradabad	Rohtak	Yeotmal
Bijapur	Faridpur	Jhelum	Multan	Saharanpur	
Bijnor	Farrukhabad	Jubbulpore	Murshidabad	Salem	
Birbhum	Fatehpur	Jullundur	Muttra	Sambalpur	
Bogra	Ferozepore	Kaira	Muzaffargarh	Santal Parganas	
Broach and Panch Mahals	Fyzabad	Kamrup	Muzaffarnagar	Saran	

Appendix 2: Details of random coefficients models

As discussed above, the general model estimated is

$$LPOP_{it} = \pi_{0i} + \pi_{1i} T_t + \pi_{2i} FLU_t + \pi_{3i} T_t FLU_t + \varepsilon_{it},$$

where *i* and *t* index districts and time in years. The coefficient estimates π_{0i}, π_{1i}, π_{2i}, and π_{3i} are defined as

$$\begin{aligned}
\pi_{0i} &= \gamma_{00} + \zeta_{0i} \\
\pi_{1i} &= \gamma_{10} + \zeta_{1i} \\
\pi_{2i} &= \gamma_{20} + \zeta_{2i} \\
\pi_{3i} &= \gamma_{30} + \zeta_{3i}
\end{aligned}$$

where it is assumed that

$$\varepsilon_{ij} \sim N(0, \sigma_{\varepsilon}^2)$$

and

$$\begin{bmatrix} \zeta_{0i} \\ \zeta_{1i} \\ \zeta_{2i} \\ \zeta_{3i} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_0^2 & \sigma_{01} & \sigma_{02} & \sigma_{03} \\ \sigma_{10} & \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{20} & \sigma_{21} & \sigma_2^2 & \sigma_{23} \\ \sigma_{30} & \sigma_{31} & \sigma_{32} & \sigma_3^2 \end{bmatrix} \right).$$

The coefficients are modeled as varying randomly across districts, and the estimates reported in Table 1 are the mean coefficients across all districts. Details of these models are provided in SAS (2011b).

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Table 1: Population Growth Models with Influenza Mortality Estimates

	MODEL SPECIFICATION			
Coefficient Estimate	Unrestricted	Restricted (Davis)†	Unrestricted without 1891	Restricted (Davis)† without 1891
Intercept (γ_{00})	13.67*** (0.04)	13.63*** (0.04)	13.71*** (0.04)	13.68*** (0.04)
Time Trend (γ_{10})	0.0039*** (0.0004)	0.0080*** (0.0003)	0.0050*** (0.0007)	0.0106*** (0.0003)
Flu Dummy (γ_{20})	-0.28*** (0.01)	-0.08*** (0.01)	-0.19*** (0.01)	-0.12*** (0.01)
Time Trend * Flu Dummy (γ_{30})	0.0081*** (0.0005)	---	0.0070*** (0.0007)	---
Number of Obs.	1194	1194	995	995
χ^2(Model)	4011.61***	3561.59***	3337.78***	3088.57***
	ESTIMATES OF KEY DEMOGRAPHIC PHENOMENA			
Influenza Mortality (millions)	11.41	15.46	13.04	23.36
Influenza Mortality (millions) Based on Equivalent Fixed Coefficients Model	11.59	16.22	13.44	24.39
Annual Population Growth Rate to 1918	0.39%	0.80%	0.50%	1.06%
Annual Population Growth Rate after 1918	1.20%	0.80%	1.20%	1.06%

Standard errors in parentheses

*** $p < 0.01$

†Population growth rates before and after 1918 are restricted to be equal

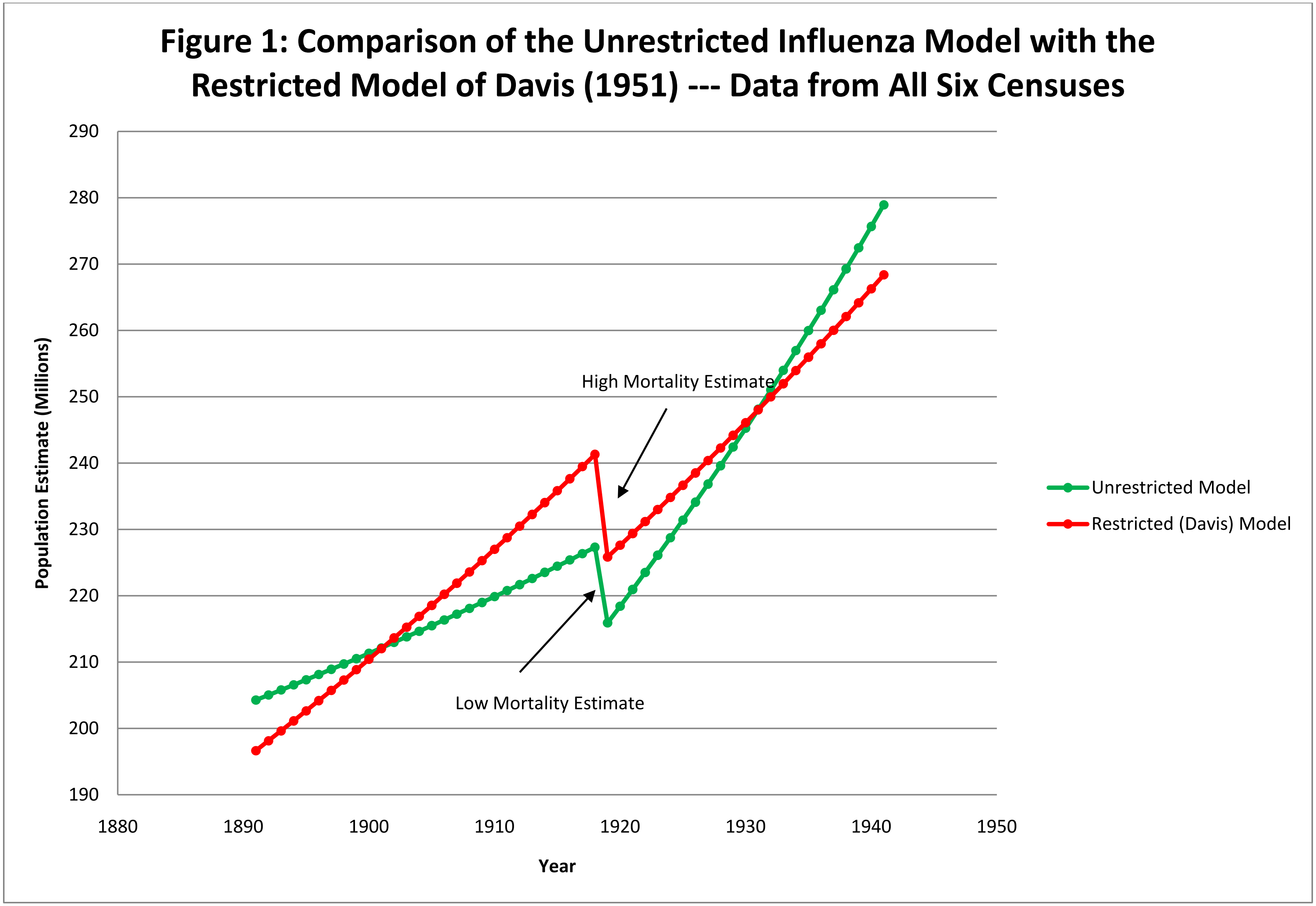
Figure 1: Comparison of the Unrestricted Influenza Model with the Restricted Model of Davis (1951) --- Data from All Six Censuses
Population Estimate (Millions)
290
280
270
260
250
240
230
220
210
200
190
1880
1890
1900
1910
1920
1930
1940
1950
Year
High Mortality Estimate
Low Mortality Estimate
Unrestricted Model
Restricted (Davis) Model

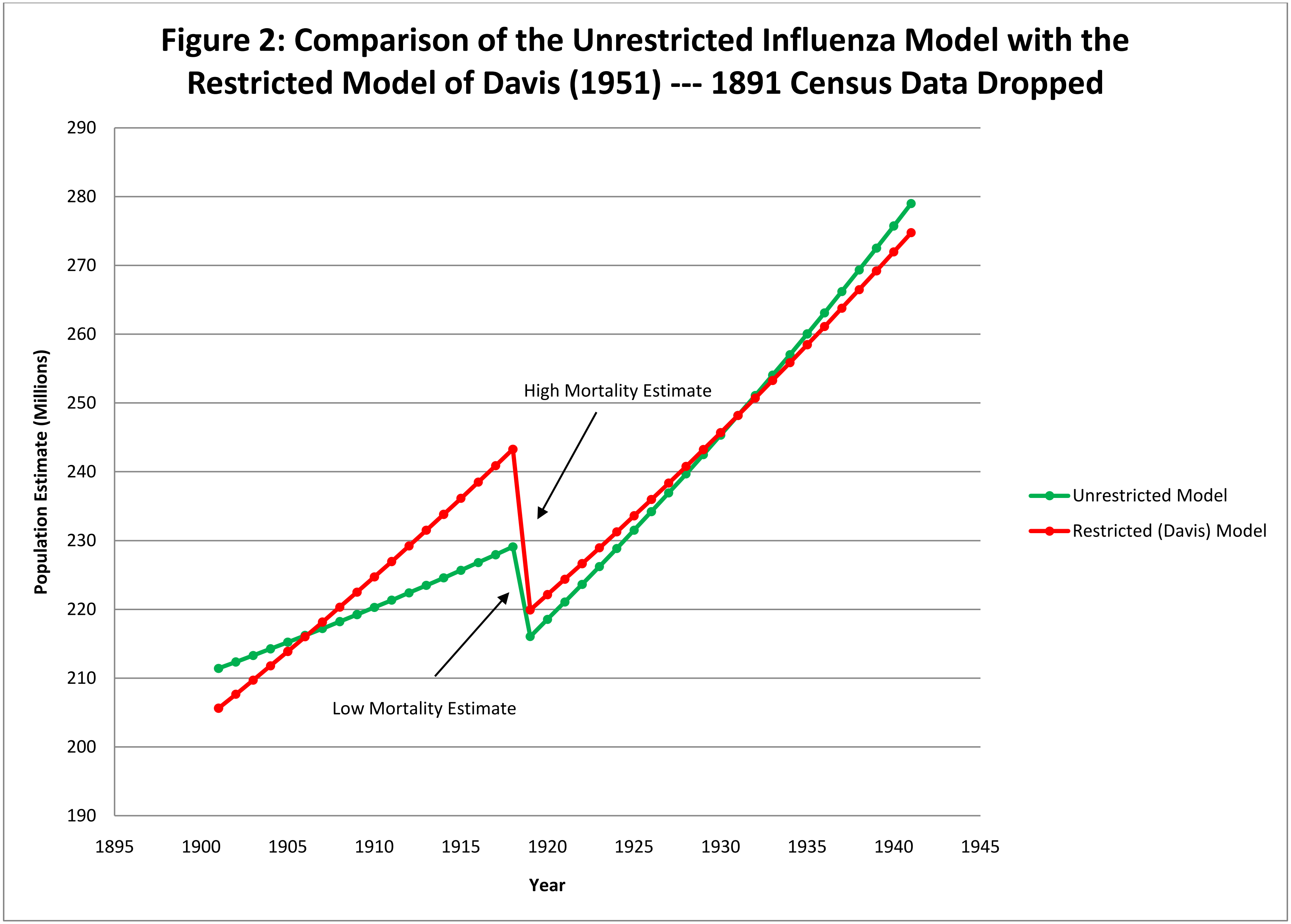
Figure 2: Comparison of the Unrestricted Influenza Model with the Restricted Model of Davis (1951) --- 1891 Census Data Dropped
Population Estimate (Millions)
290
280
270
260
250
240
230
220
210
200
190
1895
1900
1905
1910
1915
1920
1925
1930
1935
1940
1945
Year
High Mortality Estimate
Low Mortality Estimate
Unrestricted Model
Restricted (Davis) Model